



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2023	M.TECH Dual Degree 5Y	IIT Kharagpur	9.09 / 10
2018	Indian School Certificate	Hill Top School	94%
2016	Indian Certificate of Secondary Education	Hill Top School	96.8%

INTERNSHIPS

Deep Learning and Software Development Intern | UCL London

Apr'21 - Present

- Pioneered a platform to integrate effects of influential content from Twitter, FT News and TikTok with time-series stock data of companies
- Developed the back-end of a website for **ECFRI** (European Council for Financial Research and Innovation) using Django framework
- Established a platform that would serve **20+** universities to develop algorithms using **NLP** for targeted company sentiment analysis
- Scrapped **1000+** tweets using snsrape and generated sentence embeddings using **FinBert**, **SBert**, **GloVe** and **Google Universal Encoder**
- Engineered an **LSTM** Model to acquire an enhanced R2 score of **0.98** and RMSE of **2.34** using Technical Analysis on S&P 500 ES Data

NLP and Software Development Intern | Algozenith | Nagpur

Mar'21 - May'21

- Executed an Applicant Tracking System for the company and an efficient framework to generate **1000+** CVs in PDF format automatically
- Designed a **Resume Parser** and extracted **23** varied fields including Name, Emails etc. from **500+** CVs, thereby saving **1000+** person-hours
- Conceptualized a **Linear Time** algorithm to extract sections comprising Experience, Achievements, Publications, Skills and Coursework
- Procured **1000+** professional and technical resumes within seconds after transforming user entered information in JSON using NLP

PROJECTS

Deep Learning Project | University of Montreal | Canada

May'20 - Jul'20

- Conserved **100+** hours of manual cleaning of MEG Data by designing a deep learning architecture based on LFADS on ComputeCanada
- Implemented a non-linear **Recurrent Neural Network-based autoencoder** framework which widely applies to the field of MEG research
- Yielded a refined training and validation loss of **0.005** by conducting **5-Fold** cross-validation on **Graham** server using **20** subjects' data
- Preprocessed and converted Magnetoencephalography(MEG) data from .fif format to NumPy arrays utilising the MNE-Python toolkit

Computer-Supported Collaborative Case Study Analysis | IIM Kozhikode

Apr'20 - Aug'20

- Built a live platform to quantitatively analyse online group discussions on Whatsapp and produce individual performance scores instantly
- Scrutinised **50+** Whatsapp Discussion text files to produce **10** performance features using NLTK and RE, thereby saving **100+** person-hours
- Incorporated features like **Sentiment** Analyser, **Emoticon** Extractor, Message **Classifier**, **Wordcloud**, and Extractive **Summarization**
- Generated **200+** analysis reports within minutes highlighting the Grading System built on pedagogical performance parameters

COMPETITION/CONFERENCE

Runners-up | Intel AI Entrepre-Neural - A Deep Learning Competition

Apr'21

- Modeled **Autismo** (as a Web App) - A Toddler Autism detection model using both facial images and a screening test for diagnosing Autism
- Obtained an accuracy of **98%** and **88%** by deploying **RFC** for classification using question set and by using **VGGFace** Model, respectively
- Reduced the total inference time from **61.77** seconds to **45.764** seconds and improved autistic recall from **0.83** to **0.92** by using **OpenVino**

Runners-up | Data Analytics Competition | Institute Wellness Group

Nov'20

- Administered and conceptualised a **lexicon-based text classifier** for the binary classification of comments into bully or non-bully
- Boosted accuracy to **98.01%** using **MLP** outperforming **97.75%** and **97.95%** obtained from **Random Forest** and **Linear SVM**, respectively

AWARDS AND ACHIEVEMENTS

- **Competitive Coding** - Google KickStart Round C - Ranked **351** | CodeChef Rating - **1808 (4 stars)** | CodeForces Rating - **1485 (Specialist)**
- Bagged **Zonal Rank - 2** and **International Rank - 92** in the International Mathematics Olympiad 2018 by Science Olympiad Foundation
- Adjudged the **best All-Round** student of Class 12 and received the **TATA Hitachi Award** by TATA MOTORS Limited, Jamshedpur
- Attained **Zonal Rank - 3** and **International Rank - 77** in the National Science Olympiad 2018 organised by Science Olympiad Foundation
- Amongst the top **10%** students of **1400** first years to get a **Department Change** from Manufacturing to Chemical Engineering

COURSEWORK INFORMATION

Academic Courses: Programming and Data Structures | Differential Calculus | Data Analytics | Algorithms and Data Structures | Deep Learning Foundations and Applications | Probability and Statistics | Integral Calculus | Transform Calculus | **MOOCs:** Hands-on Machine Learning | Machine Learning | Deep Learning specialisation | Probability and Applied Statistics (MIT Courseware) | AlgoZenith (Algorithms)

POSITIONS OF RESPONSIBILITY

Design and Media Head | Spring Fest 2021

Jun'20 - Mar'21

- Ensured delivery of digital and print media for publicity in **800+** colleges, **550+** corporates, and a media reach of **2,70,000+** subscribers
- **Co-led a 2-tiered** design team of **17** members; designed **20+** digital posters, theme for website and oversaw the creativity of **70+** other
- Created a two-themed website for the online fest which subsequently generated over **2,00,000** views and **25,000** registrations

Editor | Awaaz | IIT Kharagpur

Jun'20 - Jun'21

- Pioneered the first coloured newsletter design and conducted **50+** fundraisers aimed at raising INR **1 Crore+** during the Covid pandemic
- Developed and published insights regarding issues like mess problems, elections, placements, internships, and general championship
- Led a team of more than **25** members to manage the publication of the monthly newsletter along with the conduction of various events

SKILLS AND EXPERTISE

Programming Languages: C++ | Python | C | BlueJ | **Libraries and Tools:** Scikit-Learn | Pandas | Latex | Numpy | PyTorch | TensorFlow | Keras | NLTK | Spacy | RE | OpenCv | Matplotlib | SeaBorn | MNE-Python | Django | PDFMiner | PyPanDoc | GitHub | Tableau | SSH